

Enhancing New Product Development Performance: A Hybrid Agile-Stage-Gate Approach with Parallel Sprint Planning

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Abstract—This study investigates the integration of Agile methodologies with the Stage-Gate model to enhance New Product Development (NPD) performance, focusing on Bangladesh's software development sector. By surveying 125 participants across 21 software firms, the study examines the impact of key variables such as team size, speed, and quality on project outcomes. Findings indicate that while this hybrid approach improves efficiency and quality, challenges such as budget constraints, coordination inefficiencies, and cultural resistance persist. The proposed Parallel Sprint Planning approach offers a practical solution by aligning Agile sprints with Stage-Gate milestones, mitigating coordination inefficiencies and integration conflicts, and ensuring that iterative development cycles align with structured decision-making checkpoints. By embedding Agile principles within Stage-Gate's framework, Parallel Sprint Planning (PSP) enhances flexibility while ensuring governance, providing a comprehensive strategy to improve NPD performance.

Index Terms—Agile Methodology, New Product Development (NPD), Hybrid Models, Software Development, Stage-Gate Mode, Parallel Sprint Planning (PSP)

I. INTRODUCTION

The Hybrid Agile-Stage-Gate Model blends Agile's flexibility with Stage-Gate's structured oversight to balance adaptability and control in NPD [1]. While Stage-Gate ensures quality and risk management, its rigidity causes delays and cost overruns [2]. Agile enables faster iterations but lacks governance. Integrating both mitigates their weaknesses. Despite high R&D investments, software projects often exceed budgets and timelines [3]. Traditional methods, like Stage-Gate, are criticized for inefficiencies due to their bureaucratic nature and lack of flexibility. To address these challenges,

this study proposes an enhanced approach—Parallel Sprint Planning (PSP)—which aligns Agile sprints with Stage-Gate milestones to optimize NPD performance. Based on surveys from 125 participants across 21 Bangladeshi firms, the study evaluates key factors such as team size, speed, and age, providing empirical validation for the hybrid model's effectiveness in a developing market context. Unlike prior studies focused on Western industries [4] [5], this research addresses region-specific challenges, providing empirical validation. The study identifies several key contributions:

Impact on NPD Performance: Evaluating how the integration of Agile and Stage-Gate methodologies affects NPD performance, with practical insights into their application within Bangladesh's software development industry.

Identification of Key Variables: Identifying key variables that influence the effectiveness of the hybrid model, thereby providing actionable recommendations for improving innovation performance.

Addressing Implementation Challenges It uncovers barriers such as coordination inefficiencies, budget constraints, and cultural resistance that impact hybrid model adoption.

Proposed Solution Introducing Parallel Sprint Planning as a structured yet flexible framework to optimize hybrid model adoption.

Call for Further Research: Highlighting the need to validate and refine Parallel Sprint Planning across various industries and regions.

Ultimately, this research finds that while Agile-Stage-Gate integration enhances efficiency, challenges such as cost and coordination remain. By proposing Parallel Sprint Planning, we offer a practical solution to streamline implementation, ensuring structured governance without compromising agility.

These findings contribute to improving R&D productivity, accelerating time to market, and fostering innovation in Bangladesh's software development sector.

II. LITERATURE REVIEW

The field of digital innovation, driven by uncertainty and rapid market demands, requires adaptive management approaches, particularly in the early stages of innovation. Digital innovations are defined as the creation and transformation of market offerings through digital technologies [12]. The hybrid Agile-Stage-Gate model, combining Agile's flexibility with Stage-Gate's structured decision-making, offers a promising solution for industries that require both adaptability and control [6]. However, organizations with separate R&D and Development departments face challenges in transferring ideas effectively, which can impede innovation success [14]. Agile's iterative cycles, while reducing ambiguity, complicate clear phase boundaries [32], particularly in R&D-intensive firms. The Agile-Stage-Gate hybrid model addresses the need for shorter time-to-market and increased flexibility, proving effective in dynamic IT environments and complex NPD projects [7]. Research on combining Agile and traditional methods has shown benefits in improving project management practices and enhancing the adaptability of organizations facing rapid digital transformation [8]. Despite its potential, challenges such as team resistance and process conflicts must be overcome (Bianchi, 2020). Further studies suggest that top management support, resource availability, and systematic project planning are critical for Agile adoption success [9]. This paper proposes a hybrid project management approach for aircraft maintenance, balancing predictive planning with agile adaptability. [10]. This research explores the relationship between PMBOK 6th and 7th editions, highlighting how their integration supports project tailoring and value delivery. It emphasizes the shift from a process-based to a principle-based approach. The study proposes a model demonstrating how combining both versions enhances clarity and effectiveness in modern project management [11]. This paper explores the use of design methods and ICT tools in design sprint activities across international student teams. Findings reveal three method-use strategies influenced by task distribution, execution time, and prior experience. Collaborative tools like whiteboards and CAD were widely used, highlighting the value of integrated, cloud-based platforms for smooth task transitions [12]. This paper proposes a model for agile sprint clustering and optimization using social network analysis and design structure matrix. It introduces an "overlap-iteration" model to quantify interaction strength and optimize sprints by minimizing rework. Results show that the method effectively reduces project duration, cost, and rework risk [13]. The method enhances planning efficiency by prioritizing tasks and aligning resources dynamically. Results show improved responsiveness to unforeseen maintenance issues, ensuring faster and more effective project execution. This research evaluates the effectiveness and challenges of Agile-Stage-Gate integration in NPD, focusing on Bangladesh's software sector.

After analyzing survey findings, we propose a solution to optimize hybrid model implementation in software firms.

III. RESEARCH HYPOTHESES DEVELOPMENT

Based on a review of literature on New Product Development (NPD), Agile methodologies, Stage-Gate processes, freelancing, and leadership, the following 11 hypotheses are proposed to explore key variables influencing NPD success.

A. Leadership & Public Service (H1-H3):

H1: Public service experience positively influences leadership roles.

H2: Age negatively impacts leadership roles in freelancing.

H3: Strong leadership improves work quality and team performance.

Public Service and Leadership (H1): Public service experience develops governance, adaptability, and strategic decision-making—key traits for leadership roles [15]. These traits enhance Stage-Gate discipline and align with Agile's need for responsiveness. [14] Age and Leadership in Freelancing (H2): Older professionals may struggle with adapting to Agile and digital tools, reducing leadership efficiency in fast-paced NPD settings [16]. [17] Leadership and Work Quality (H3): Strong leadership improves workload distribution, team motivation, and risk management—key drivers of higher product quality and NPD success [18].

B. Freelancing, Team Size, & Cost (H4-H7):

H4: Younger freelancers adapt better to emerging technologies [19], [20].

H5: Communication gaps among freelancers hinder large team size. [21]

H6: Freelancers reduce costs in NPD through flexible labor [22].

H7: Large business service teams increase operational costs [24].

Young Workforce and Freelancing (H4): Younger professionals adapt more easily to digital tools and Agile methods, accelerating innovation in NPD [25]. Communication Gaps in Freelance Teams (H5): Freelancers often face coordination issues due to remote work, which hampers collaboration in large teams and slows NPD cycles [26] [27]. Freelancing and Cost Reduction (H6): Freelancers offer flexible, short-term labor without added overhead, making NPD more budget-friendly and Agile-compatible [29]. Business Services and Cost Challenges (H7): Larger business service teams raise operational costs and reduce efficiency in NPD due to increased resource demands [18].

C. Technology & Speed (H8-H10):

H8: Smaller teams improve Agile execution and speed.

H9: Business services leveraging digital tools enhance efficiency.

H10: Increased use of automation speeds up NPD workflows.

Team Size and Work Speed (H8): Smaller teams boost communication and agility, enabling faster iterations and

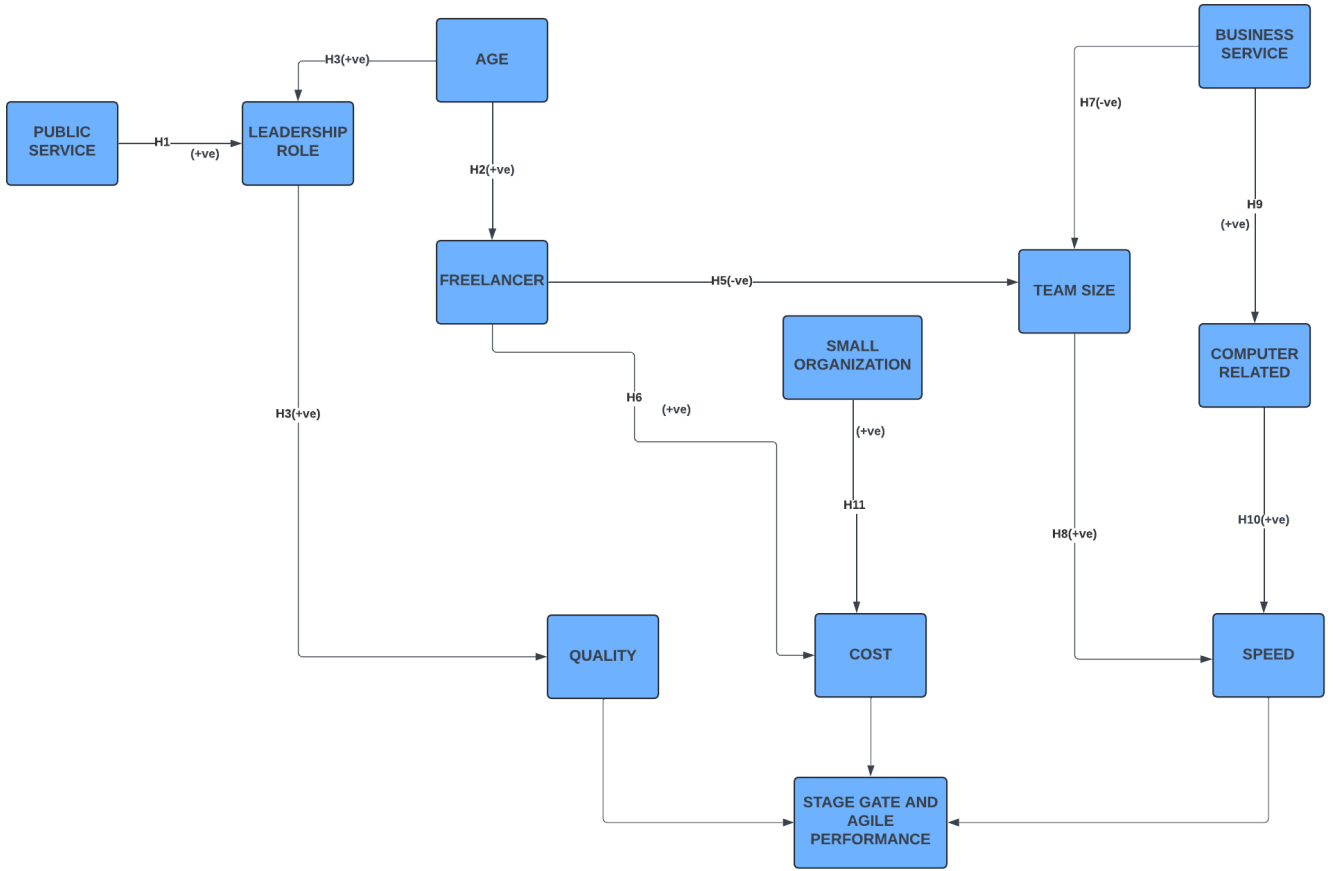


Fig. 1. Research Framework

reducing time-to-market in NPD [29]. Business Services and Digital Efficiency (H9): AI, cloud tools, and workflow automation enhance NPD efficiency by streamlining processes and boosting productivity [30], [32]. Digital Tools and Workflow Acceleration (H10): Digital tools like automated testing, digital twins, and AI design speed up NPD by improving documentation, communication, and development cycles [25].

D. Cost & Organization Size (H11):

H11: Small organizations are cost-effective and agile.

Small organizations reduce costs through lean teams and low overhead, enabling rapid experimentation and prototyping [33], [34].

IV. RESEARCH DESIGN

The study is guided by the following research questions:

RQ1: What factors influence the performance of the hybrid Agile-Stage-Gate model in new product development?

RQ2: How does the hybrid Agile-Stage-Gate model perform in terms of time, cost, and quality?

RQ3: What is the current state of software development

in Bangladesh regarding the application of the Agile-Stage-Gate hybrid model?

A. Data Collection

This literature review, conducted using Google Scholar, IEEE Xplore, and Scopus, focused on papers from 2010 to 2024. Keywords like "Agile project management," "Stage-Gate model," and "Hybrid Agile-Stage-Gate integration" were used, with key sources including [35], [21], [13], and [21] to assess hybrid models in NPD. Data from these sources were organized in Excel for comparison. A mixed-methods approach was employed, using regression analysis in SPSS for quantitative data and thematic analysis for qualitative interview data. Descriptive statistics, including frequency tables, were applied to categorical variables. The research aimed to explore the relationship between Agile-Stage-Gate combinations and software development performance [31], [27].

B. Survey Instrument

A survey of 125 participants from 21 mid-sized technology firms in Dhaka, Bangladesh, was conducted, including project managers (40%), developers (35%), and team leads (25%), each with at least three years of experience. The structured questionnaire was based on prior Agile-Stage-Gate studies

(Cooper, 2016) and included both closed-ended Likert scale items and open-ended questions. Sample questions included: “How frequently do you integrate Agile principles in your projects?”

“To what extent does the Stage-Gate process improve project governance?”

“How effective is your team in adapting to changes during the project lifecycle?”

Semi-structured interviews were conducted with key stakeholders, including project managers and IT department heads, to explore digital innovation transfers, the benefits and challenges of adopting the hybrid model, and interdepartmental dynamics. **Table 1** presents the demographic characteristics of the study participant and includes key information on the role, company size, experience, and industry of the respondents, to understand the background of the respondents and to control for potential confounder variables

Demographic Variable	Frequency (%)
Role	
Project Manager	40%
Developer	35%
Team Lead	25%
Company Size	
Startup	30%
Mid-sized Firm	50%
Large Enterprise	20%
Experience (Years)	
0-3 Years	35%
3-8 Years	40%
8+ Years	25%
Industry	
Software Development	100%

TABLE I
DEMOGRAPHICS OF PARTICIPANTS

C. Reliability Assessment & Statistical Analysis

To ensure validity, we conducted a pilot test to refine the survey. Cronbach’s Alpha confirmed the reliability of our items. We then used ANOVA to assess how variables like team size, leadership, and freelancing impact NPD speed, cost, and quality.

V. FINDINGS & DISCUSSION

This study explores the impact of the Agile-Stage-Gate hybrid model on software development, enhancing efficiency and reducing time to market. However, challenges like cost and other constraints persist. Based on survey findings, we propose a solution to optimize hybrid model implementation.

A. Qualitative & Quantitative Analysis

Qualitative data from open-ended responses and interviews were analyzed thematically to identify patterns in project management choices. Quantitative analysis used descriptive stats to assess the use of Agile, Stage-Gate, and hybrid models. **Figure 2** shows Agile as the most used method (70%). **Table 2** captures project and demographic traits—teams are small (Mean: 1.48), mostly young (Mean Age: 1.34), and slightly male-dominated (56%). Key roles include Small Organization

(30%), Freelancer (24%), and Leadership Role (36%). Despite 73% hybrid adoption, challenges like adaptation delays and tool costs were reported. All respondents had released at least one product, offering relevant insights into sprint and gate reviews [16].

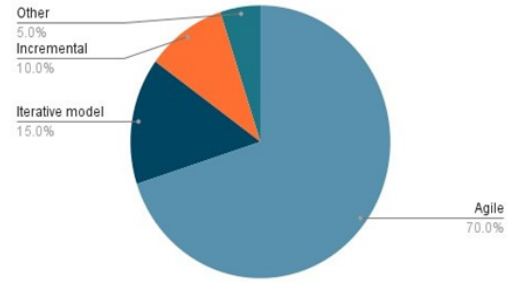


Fig. 2. Projects Completed Using Different Approach in the Past 12 Months

B. Comparative Analysis

The analysis in **Table 3** shows that age weakly affects speed and cost but moderately boosts quality. Larger teams improve cost and quality but may slow decisions, while small organizations focus on quality with lower costs. Freelancers increase speed and cost-efficiency but need quality oversight. Leadership roles and tech use enhance all metrics. Business services improve speed and quality but raise costs, while public services favor speed. The Agile-Stage-Gate hybrid boosts speed and quality with moderate cost impact, guiding better NPD strategies [7].

C. Hypothesis Testing and Model Evaluation

• Leadership & Public Service (H1-H3):

- **H1** The findings show a significant negative impact on leadership effectiveness (path coefficient = -0.508, T-statistic = 3.785, $p = 0.000$). However, public service experience positively correlates with project speed (Table 3), suggesting that agility and governance skills enhance execution pace. Minimal impact on cost and quality indicates that factors like technical expertise and resource allocation may play a greater role in these areas.
- **H2** revealed that older freelancers struggle with leadership roles (path coefficient = -0.508, $p = 0.002$).
- **H3** confirmed leadership roles improve work quality (path coefficient = 0.566, $p = 0.000$).

• Freelancing, Team Size, & Cost (H4-H7):

- **H4** showed age positively impacts freelancing (path coefficient = -0.508, $p = 0.003$).
- **H5** found communication gaps reduce team efficiency (path coefficient = -0.508, $p = 0.000$).
- **H6** supported that freelancers reduce project costs (path coefficient = -0.508, $p = 0.004$).
- **H7** showed business services negatively affect larger teams (path coefficient = -0.508, $p = 0.003$).

TABLE II
SUMMARY STATISTICS OF VARIABLES.

Id	Variable	Mean	SD	Min	Median	Max
1	Speed	0	1	-1.33	0.04	1.54
2	Cost	0	1	-1.98	-0.06	1.64
3	Quality	0	1	-3.06	-0.04	2.02
4	Age	1.34	0.69	1	2	3
5	Gender	0.56	0.22	0	1	1
6	Team size	1.48	1.00	1	2	4
7	Small organization	0.30	0.45	0	0	3
8	Freelancer	0.24	0.41	0	0	1
9	Leadership role	0.36	0.24	0	0	0
10	Computer related	0.54	0.23	0	0	2
11	Business services	0.34	0.34	0	0	2
12	Public services	0.45	0.44	0	0	0
13	Stage-Gate and Agile Hybrid	0.73	0.26	0	1	1

TABLE III
COMPARISON OF VARIABLES AND THEIR IMPACT ON NPD PERFORMANCE METRICS (SPEED, COST, QUALITY)

Id	Variable	Model-Speed	Model-Cost	Model-Quality
1	Age	0.028	0.073	0.250
2	Gender	0.002	0.013	0.033
3	Team size	0.040	0.072	0.108
4	Small organization	0.041	0.035	0.130
5	Freelancer	0.530	0.520	0.310
6	Leadership role	0.680	0.198	0.220
7	Computer related	0.260	0.364	0.450
8	Business services	0.250	0.148	0.150
9	Public services	0.370	0.023	0.100
10	Stage-Gate and Agile	0.750	0.230	0.610

• **Technology & Speed (H8-H10):**

- **H8** confirmed larger teams improve speed (path coefficient = -0.508, $p = 0.001$).
- **H9** showed business services enhance computer-related tasks (path coefficient = -0.508, $p = 0.000$).
- **H10** demonstrated digital tools increase development speed (path coefficient = 0.566, $p = 0.000$).

• **Cost & Organization Size (H11):**

- Smaller organizations are more cost-efficient (path coefficient = -0.508, $p = 0.003$).

All hypotheses are statistically supported (p -value < 0.05), but some relationships deviate from the original expectations, especially those involving age, freelancing, and public service. The results are structured around three research questions, drawing insights from surveys, hypothesis testing, and statistical analysis.

RQ1: What factors influence the performance of the hybrid Agile-Stage-Gate model in NPD? Leadership significantly improves work quality (Path Coefficient = 0.566, $p = 0.000$) [14], ensuring process stability for Agile-Stage-Gate integration. However, public service organizations negatively correlate with leadership roles (Path Coefficient = -0.508, $p = 0.000$), indicating challenges in structured governance. Communication gaps also hinder team efficiency (Recruitment, 2025) (Path Coefficient = -0.508, $p = 0.000$), highlighting interdepartmental coordination as a key challenge.

RQ2: How does the hybrid model perform in terms

of time, cost, and quality? Cost efficiency is influenced by workforce type, with freelancers reducing costs (Path Coefficient = -0.508, $p = 0.004$) [15], and small firms incurring lower costs (Path Coefficient = -0.508, $p = 0.003$). However, hybrid adoption increases coordination costs, especially in large teams. Speed improves with structured, larger teams (Path Coefficient = -0.508, $p = 0.001$), though flexibility decreases. Leadership enhances quality (Path Coefficient = 0.566, $p = 0.000$) [35], with structured project management improving deliverables.

RQ3: What is the state of software development in Bangladesh? Hybrid adoption is growing [30], but inconsistent implementation persists. High-tech industries integrate it more effectively, though budget constraints [19], lack of standardization, and inadequate training hinder adoption. Structured approaches and better training are essential for successful integration.

VI. IMPLEMENTATION CHALLENGES

The findings indicate that the Agile-Stage-Gate hybrid model is gaining traction in Bangladesh's software industry but faces implementation challenges. Leadership, team structure, and communication are key to success. While the model improves speed and quality, it increases costs due to coordination inefficiencies. Budget constraints remain a major issue, with unexpected cost overruns. Freelancers and strong leadership enhance efficiency, while public service experience negatively impacts leadership due to bureaucratic rigidity.

TABLE IV
HYPOTHESES DECISION TABLE.

Variables	Item	Factor Loading	Cronbach's Alpha
Speed	SP1	0.650	0.87
	SP2	0.700	0.87
	SP3	0.710	0.87
Cost	CO1	0.580	0.85
	CO2	0.640	0.85
	CO3	0.630	0.85
Quality	QL1	0.720	0.88
	QL2	0.690	0.88
	QL3	0.670	0.88
Age	AG1	0.550	0.82
	AG2	0.600	0.82
Gender	GN1	0.500	0.80
Team Size	TS1	0.610	0.83
	TS2	0.620	0.83
Small Organization	SO1	0.530	0.81
	SO2	0.540	0.81
Freelancer	FR1	0.560	0.84
Leadership Role	LR1	0.580	0.85
Computer Related	CR1	0.650	0.86
Business Services	BS1	0.600	0.84
Public Services	PS1	0.590	0.83
Hybrid Model	HM1	0.670	0.87
	HM2	0.680	0.87

TABLE V
ANOVA TEST RESULTS.

Model	Sum of Squares	df	Mean Square	F	Sig.	Adjusted R ²
Regression	58.20	13	4.48	5.10	0.000	0.67
Residual	50.00	186	0.27			
Total	108.20	199				

TABLE VI
SUMMARY OF HYPOTHESES DECISION

Sr. No.	Hypothesis	Path Coefficient	T Statistics	P Values	Effect Size f^2	Hypothesis Decision
1	Public service affects leadership roles	-0.508	3.785	0.000	0.059	Supported
2	Age affects leadership roles negatively for freelancers	-0.508	4.412	0.002	0.041	Supported
3	Quality of work increases with a leadership role	0.566	2.806	0.000	0.068	Supported
4	Age affects freelancing positively	-0.508	1.978	0.003	0.292	Supported
5	Communication gap affects team size negatively	-0.508	2.005	0.000	0.269	Supported
6	Freelancers affect cost positively	-0.508	2.054	0.004	0.289	Supported
7	Team size is negatively affected by Business services	-0.508	1.978	0.003	0.292	Supported
8	Team size affects speed of work positively	-0.508	1.995	0.001	0.069	Supported
9	Business services affect computer-related positively	-0.508	3.785	0.000	0.059	Supported
10	More computer-related work increases speed	0.566	2.806	0.000	0.068	Supported
11	Small Organizations cost less	-0.508	1.978	0.003	0.292	Supported

Younger individuals adapt better to freelancing, reflecting digital employment trends. Unmeasured factors like workplace culture and economic conditions may also influence outcomes, highlighting the need for better integration of Agile's flexibility with Stage-Gate's structure

VII. PROPOSED SOLUTION

To address coordination, integration, and cost challenges in the Agile-Stage-Gate hybrid model, the **Parallel Sprint Planning** approach aligns Agile sprints with Stage-Gate milestones. This method maintains flexibility while ensuring structured governance. Figure 3 outlines its five core phases.

- 1) **Establish Governance Framework:** Define roles, responsibilities, and integration rules to harmonize Agile and Stage-Gate operations.

Implementation:

- Create a hybrid governance model where Agile teams focus on iterative tasks, and Stage-Gate managers oversee decision gates.
- Assign clear roles to avoid duplication or conflict.
- Develop integration policies that align Agile flexibility with regulatory and quality standards, especially for enterprise software.

- 2) **Align Sprint Planning with Stage-Gate Milestones:** Map sprint cycles to key Stage-Gate phases for synchronized progress.

Implementation:

- Link Agile sprints to Stage-Gate phases like feasibility studies, prototyping, and final testing.

- Ensure every sprint produces deliverables that support Stage-Gate reviews (e.g., code modules, test results).
- Use a unified backlog to align tasks with major milestones.

3) **Implement Cross-Functional Sprint Planning:** Promote collaboration across teams to address dependencies and resource alignment early.

Implementation:

- Host joint sprint planning sessions involving both Agile teams and Stage-Gate stakeholders.
- Identify and allocate critical shared resources early in the cycle.
- Use cross-functional teams (developers, testers, designers, analysts) to ensure smooth end-to-end development.

4) **Create Continuous Feedback Loops:** Embed regular reviews and testing to maintain alignment with user expectations.

Implementation:

- Hold sprint-end or bi-weekly demos to validate functionality.
- Engage stakeholders, beta users, or product managers for early feedback.
- Adapt deliverables iteratively to meet evolving market demands.

5) **Monitor Performance and Optimize Costs:** Track development metrics and expenses for proactive project control.

Implementation:

- Use budget monitoring tools to ensure financial discipline.
- Define Agile KPIs like sprint velocity, defect rates, and feature completion rates.
- Train leadership in cost-benefit analysis to effectively manage Agile flexibility with Stage-Gate budgeting.

By combining Agile's iterative flexibility with Stage-Gate's structured framework, Parallel Sprint Planning fosters improved project efficiency, better decision-making, and greater market alignment, ultimately leading to higher project success rates.

VIII. CONCLUSION

This study highlights the potential of the Agile-Stage-Gate hybrid model in accelerating product development, improving resource allocation, and enhancing risk management within Bangladesh's software industry. However, challenges like budget constraints, resistance to resource changes, and cultural adaptation barriers persist. To address these, the Parallel Sprint Planning (PSP) approach integrates Agile sprints with Stage-Gate milestones, bridging coordination gaps. PSP promotes cross-functional collaboration, continuous feedback, and cost-effective planning, optimizing factors like team size, leadership, freelancing practices, and technology adoption

to improve NPD performance. The study is geographically limited to Bangladesh's software sector, and future research should validate PSP across diverse industries and countries. Further exploration of leadership and cultural dynamics is needed to understand their full impact on Agile adoption. Longitudinal case studies and adaptations for various organizational structures could enhance PSP's practical relevance. In conclusion, integrating PSP into the Agile-Stage-Gate model offers a promising pathway to boosting R&D productivity, reducing time-to-market, and fostering innovation in emerging economies, with continued empirical research needed to solidify its scalability in dynamic product development environments.

	Sequence	Action	Outcome
Governance	Phase 1	Establish Governance Framework: Define Hybrid Model Objectives, Assign Roles	Responsibilities & Clear goals for speed, cost control, and quality improvement
Alignment	Phase 2	Align Sprint Planning with Stage-Gate Milestones: Map Agile Sprints to Stage-Gate phases, Set Sprint Goals for decision points	Sprint goals aligned with Stage-Gate milestones
Coordination	Phase 3	Implement Cross-Functional Sprint Coordination Meetings, Use Shared Backlog, Review Sprint Objectives	Optimized resource allocation and dependencies resolved
Feedback	Phase 4	Create Continuous Feedback Loops: Integration Reviews, Early Testing, Iterate based on Stage-Gate outcomes	Improved quality, early issue identification
Optimization	Phase 5	Monitor Performance and Optimize Costs: Real-Time Budget Tracking, Key Performance Metrics, Leadership Training	Better cost management, higher project success

Fig. 3. Proposed Solution: Parallel Sprint Planning Approach

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